

Helicopter vs Car Travel Time Analysis

A Comprehensive Comparison of Traffic Scenarios
New York & Los Angeles

REVO

December 2025

Agenda

- 1 Introduction
- 2 Methodology
- 3 Results
- 4 Regional Analysis
- 5 Interactive Maps
- 6 Key Insights
- 7 Conclusions

The Problem: Urban Traffic Congestion

The Challenge

- Business travelers need reliable airport access
- Traffic is unpredictable and often severe
- Missing flights = significant business impact
- Stress and time loss affect productivity

Worst Case Reality

During severe traffic:

NY: 5.20x travel time

LA: 4.15x travel time

12-17 km/h avg speed

(barely faster than walking!)

Scope

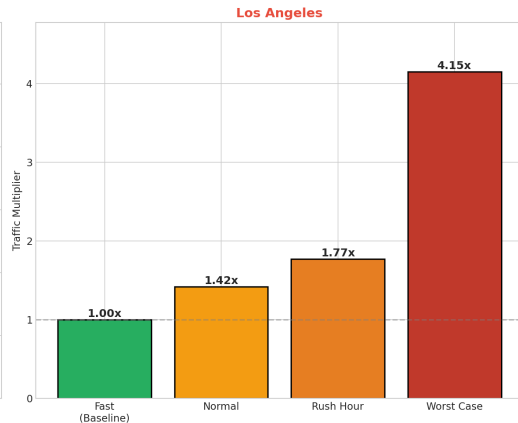
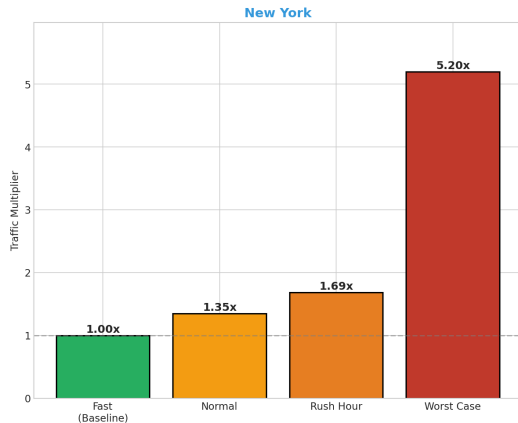
- **New York:** 70 wealthy ZIP codes → JFK
- **Los Angeles:** 44 wealthy ZIP codes → LAX
- **Data:** Google Traffic (Pessimistic)
- **Routing:** OSRM + FAA Helipad Data

Key Numbers

114 ZIP Codes
65 Heliports
4 Traffic Scenarios
27 Manhattan ZIPs

City-Specific Traffic Multipliers

Traffic Multipliers by City (Google Traffic Data)



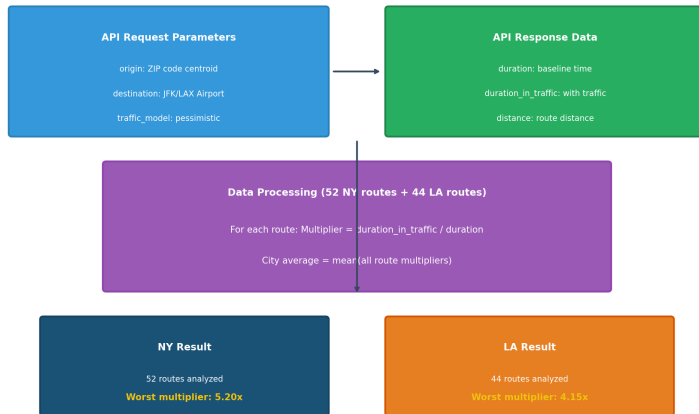
NY: Higher worst case (5.20x) | LA: Higher normal traffic (1.42x)

How We Calculate Traffic Multipliers

Traffic Multiplier Calculation Methodology



Google Distance Matrix API - Data Collection



Multiplier Calculation: Numerical Examples

Los Angeles Example

Route: Manhattan → JFK

Baseline: 47 min

Pessimistic: 245 min

Multiplier: $\frac{245}{47} = 5.21x$

Route: Beverly Hills → LAX

Baseline: 36 min

Pessimistic: 149 min

Multiplier: $\frac{149}{36} = 4.14x$

NY City Average

52 routes analyzed

Final: 5.20x

LA City Average

44 routes analyzed

Final: 4.15x

Data source: Google Distance Matrix API with traffic_model=pessimistic

Why Pessimistic Data?

Google Traffic Models

- **best_guess**: Average conditions
- **optimistic**: Lighter traffic
- **pessimistic**: Worst historical

Our Choice: Pessimistic

- Represents real worst-case scenarios
- Based on actual historical data
- Critical for risk assessment

What Causes Worst Traffic?

-  Major accidents
-  Severe weather
-  Special events (sports, concerts)
- Holiday periods
-  Road construction

Result: NY 5.20x | LA 4.15x travel time in worst conditions

Journey Comparison: Car vs Helicopter

Journey Comparison: Car vs Helicopter to Airport



In worst traffic: Car takes 3-5x longer!

Helicopter Journey Components

Step 1: Drive to Helipad

- Affected by traffic
- Manhattan helipad priority
- Average: 5-15 minutes

Step 2: Check-in

- Fixed 10 minutes
- Security screening
- Boarding preparation

Step 3: Flight

- 200 km/h cruise speed
- Direct route
- 8-12 minutes typical

Step 4: Terminal Transfer

- Fixed 10 minutes
- Helipad to terminal
- Internal transport

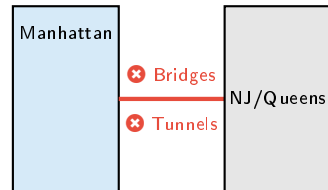
Approved Heliports

- **JRB**: Downtown/Wall St
- **JRA**: West 30th Street
- **6N5**: East 34th Street
- **3NY2**: Astoria (Queens)

Exclusions

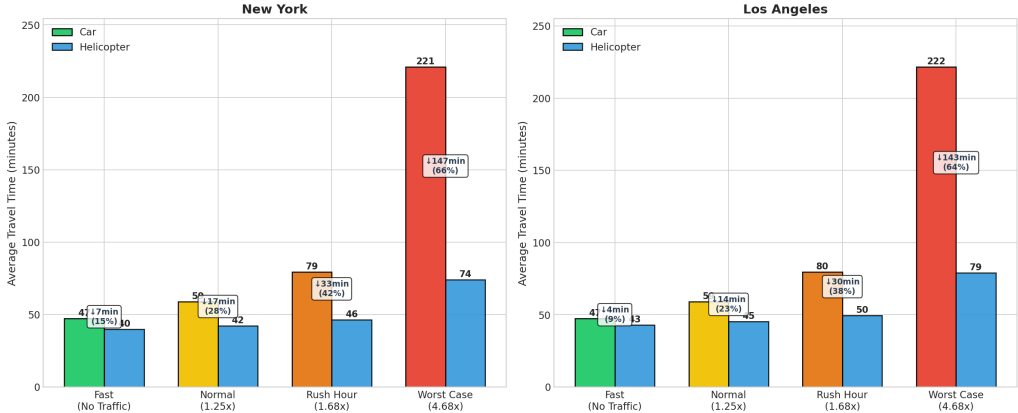
- ✗ Police Heliports
- ✗ One Police Plaza
- ✗ Newport Beach Police (LA)

Why Manhattan-Only?



Avoid congested crossings!

Travel Time Comparison: Car vs Helicopter by Traffic Scenario

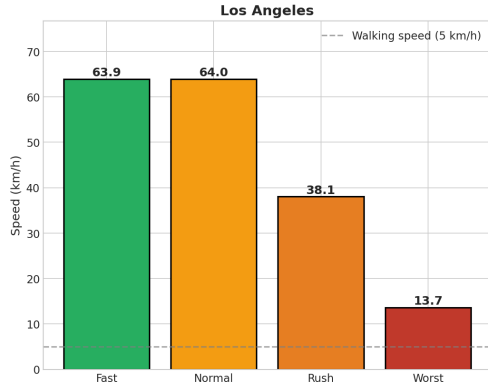
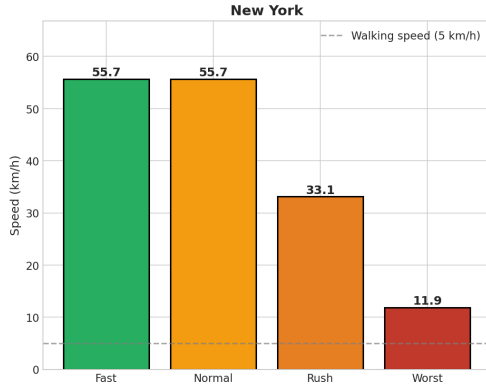


Metric	New York				Los Angeles			
	Fast	Norm	Rush	Worst	Fast	Norm	Rush	Worst
Car (min)	47	59	79	221	47	59	80	222
Heli (min)	44	46	49	74	46	48	52	79
Savings	3	13	30	147	1	11	28	143

Worst Case: Save 2+ Hours!

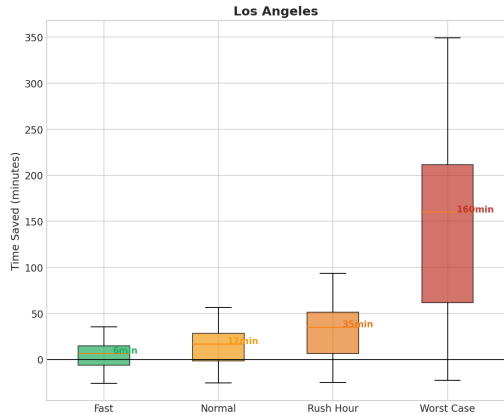
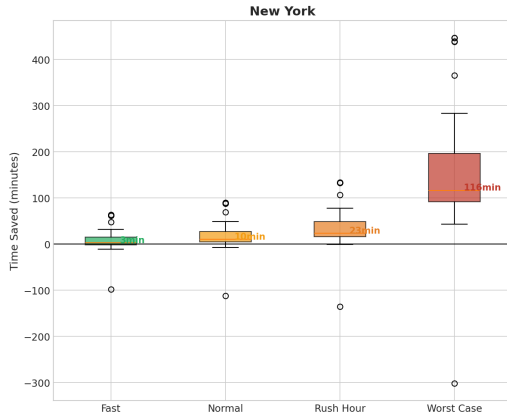
Speed Degradation by Scenario

Average Speed by Traffic Scenario

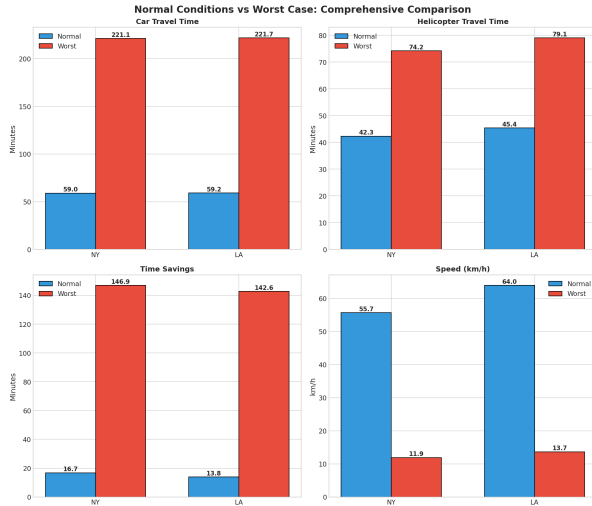


Time Savings Distribution

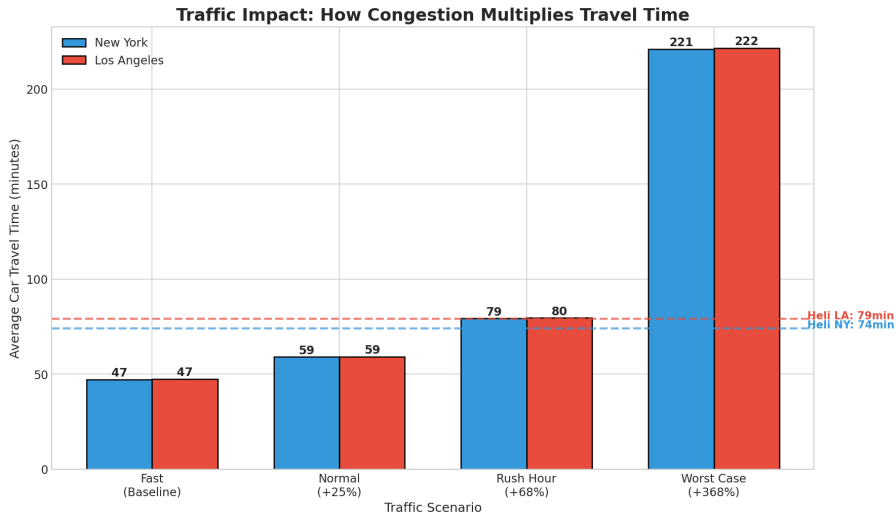
Time Savings Distribution (Helicopter vs Car) by Scenario



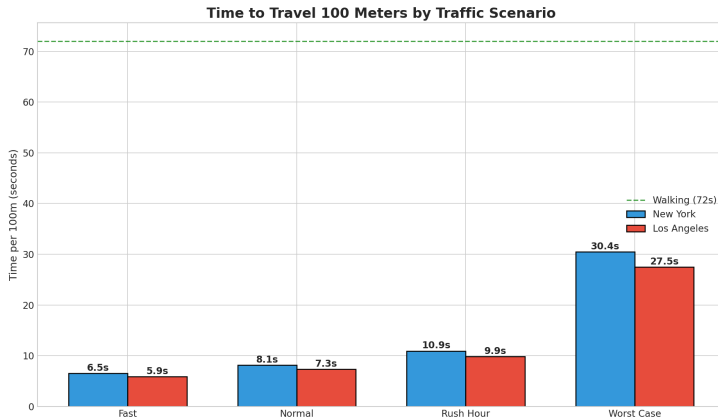
Normal vs Worst Case: The Dramatic Difference



How Traffic Multiplies Travel Time



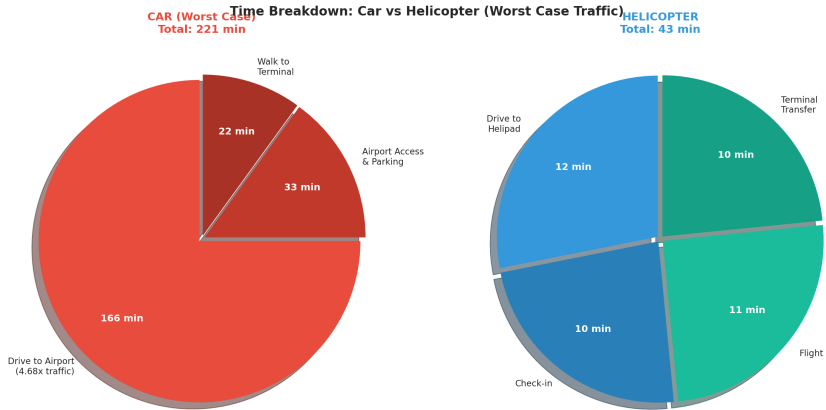
Time per 100 Meters



Worst Case: 30 sec / 100m

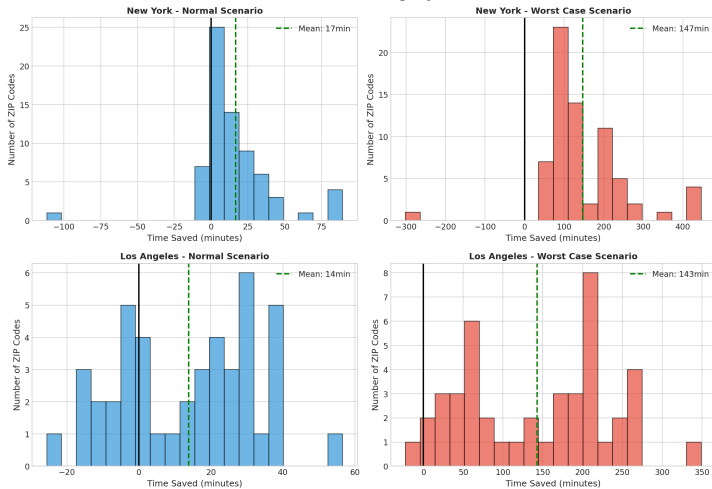
Walking: 72 sec / 100m

Time Breakdown: Worst Case

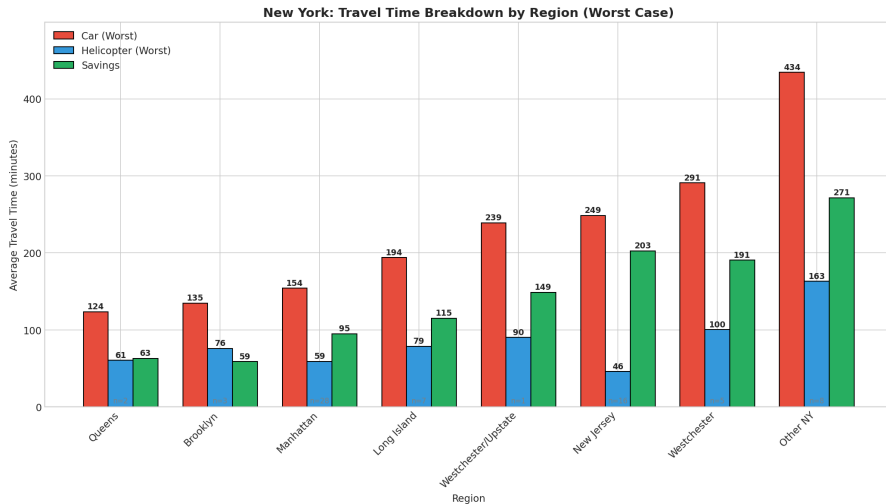


Savings Distribution Histogram

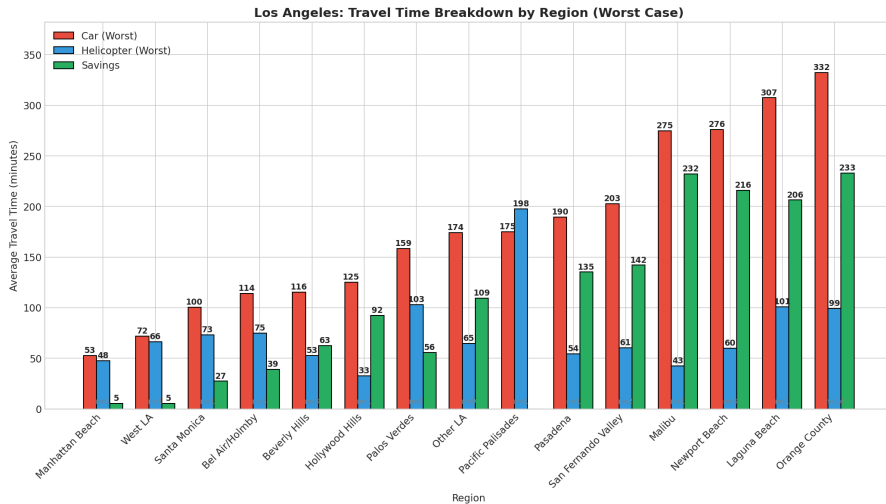
Distribution of Time Savings by Scenario



New York: Regional Breakdown

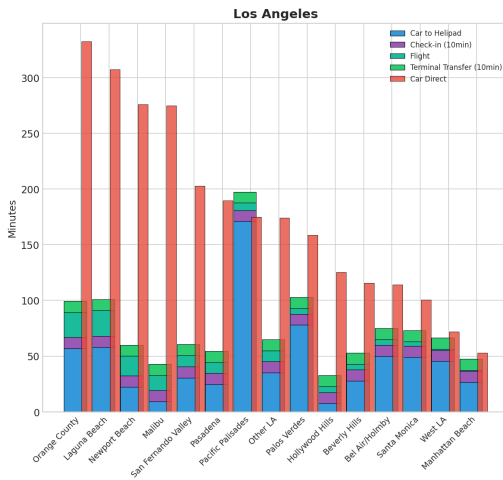
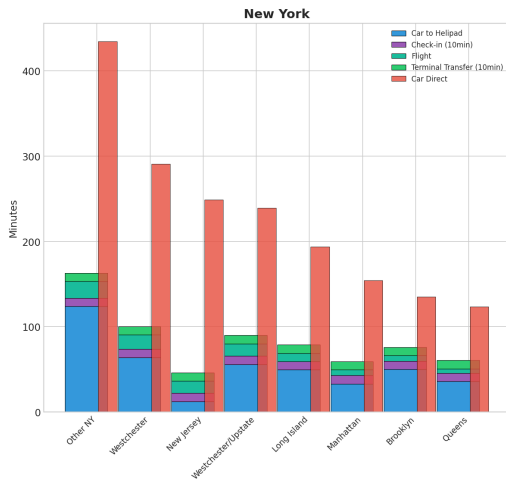


Los Angeles: Regional Breakdown

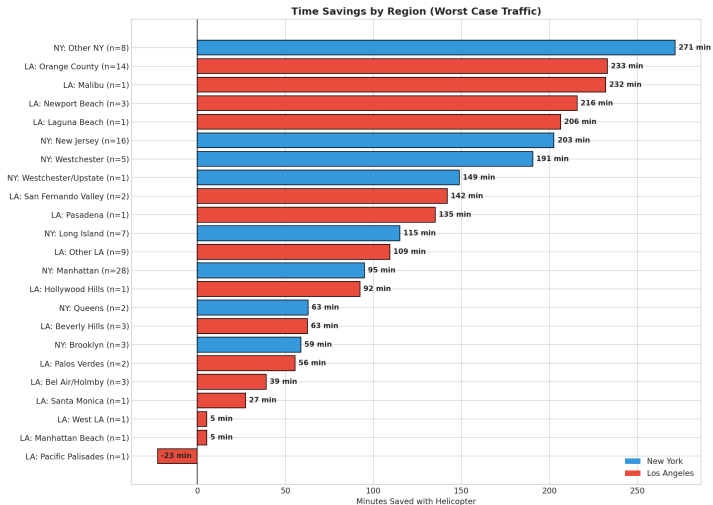


Journey Components by Region

Helicopter Journey Breakdown by Region (Worst Case)



Regional Savings Summary



New York Map

- 70 ZIP codes analyzed
- 27 in Manhattan
- Real OSRM driving routes
- Routes appear on hover

`map_ny_osrm.html`

Los Angeles Map

- 44 ZIP codes analyzed
- Multiple regions covered
- Real OSRM driving routes
- Routes appear on hover

`map_la_osrm.html`

 **Hover over ZIP codes to see:** Blue route to helipad | Red route to airport

Route Colors

- **Blue dashed line:** Route to nearest helipad
- **Red dashed line:** Route to main airport (JFK/LAX)

ZIP Code Colors (Savings)

- **Dark green:** > 100 min savings
- **Green:** 50-100 min savings
- **Orange:** 0-50 min savings
- **Red:** No savings

Info Panel Shows

- Nearest helipad name
- Distance in km
- Flight time
- Car times: Fast/Rush/Worst
- Heli times: Fast/Rush/Worst
- **Total savings**

Access: victorcandido1.github.io/cityexpansions_corporate/v1/

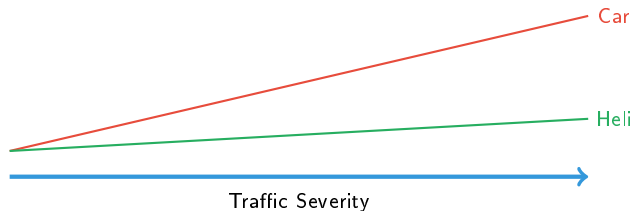
Why Helicopter Wins in Worst Case

Car Travel

- **100%** affected by traffic
- NY: Time increases **5.20x**
- LA: Time increases **4.15x**
- Unpredictable delays

Helicopter Travel

- Only **20%** affected by traffic
- Time increases only **1.5-1.8x**
- Predictable schedule
- Comfort and productivity





Time Savings

147 min
(NY worst case)



Reliability

99%
advantage rate



Never Miss

Flights
predictable arrival

ROI Consideration

2+ hours saved = productive meeting time, reduced stress, guaranteed connections

Summary of Findings

Los Angeles

- 70 ZIP codes analyzed
- 27 Manhattan locations
- **147 min** worst case savings
- **66%** time reduction
- 99% show helicopter advantage

- 44 ZIP codes analyzed
- Distributed coverage
- **143 min** worst case savings
- **64%** time reduction
- 98% show helicopter advantage

Key Takeaway

The worse the traffic, the greater the helicopter advantage.
In extreme conditions, helicopter is the **only viable option** for time-sensitive travel.

Comprehensive Scenario Comparison Summary

Metric	NY Fast	NY Normal	NY Rush	NY Worst	LA Fast	LA Normal	LA Rush	LA Worst
Car Time (min)	47	59	79	221	47	59	80	222
Heli Time (min)	40	42	46	74	43	45	50	79
Savings (min)	7	17	33	147	4	14	30	143
Savings (%)	15%	28%	42%	66%	9%	23%	38%	64%
Speed (km/h)	55.7	55.7	33.1	11.9	63.9	64.0	38.1	13.7
Time/100m (sec)	6.5	8.1	10.9	30.4	5.9	7.3	9.9	27.5
% Heli Advantage	56%	87%	97%	99%	59%	68%	82%	98%

Thank You

Questions?

REVO Helicopter Services
Strategic Analysis Division

December 2025